

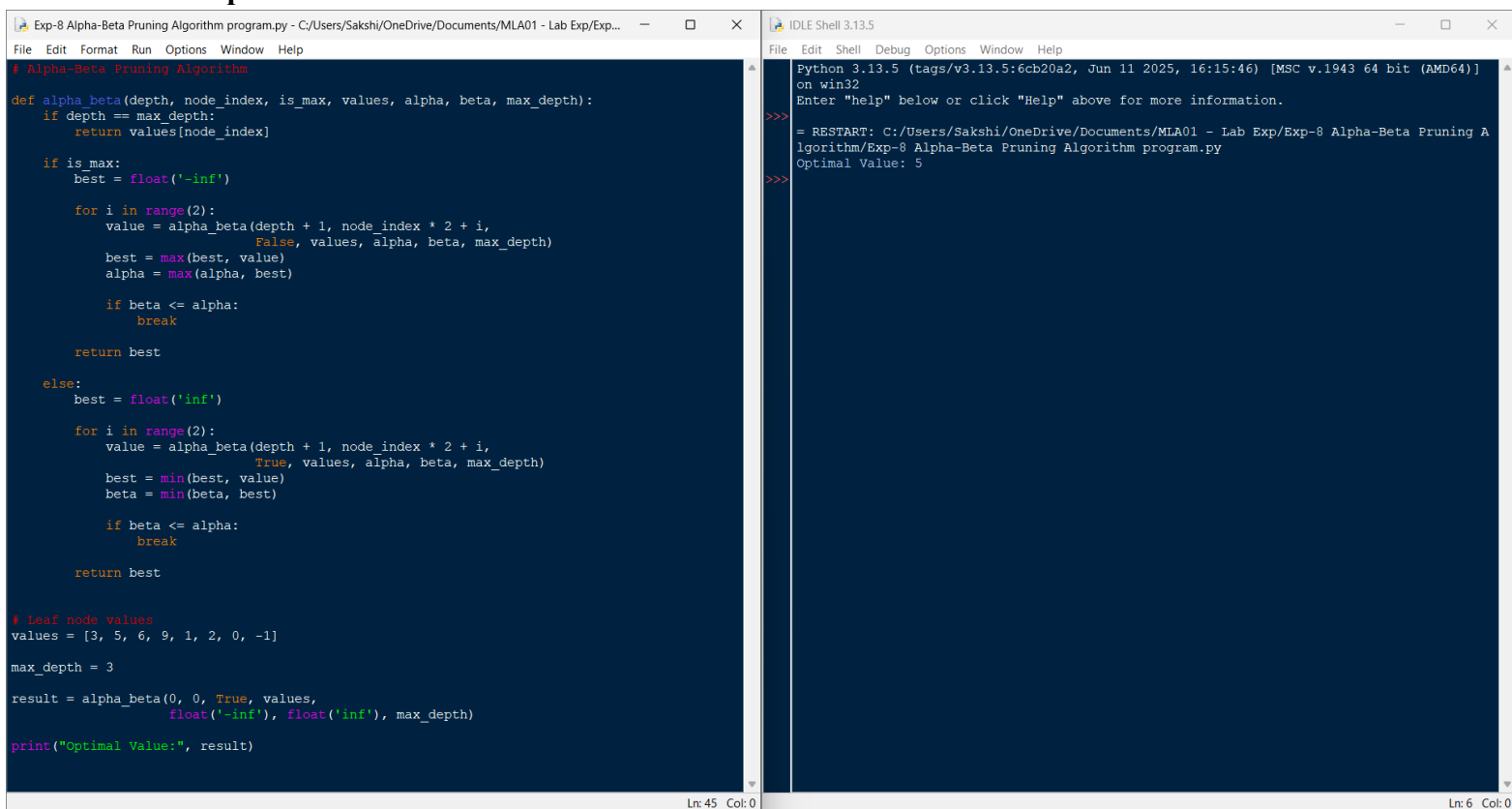
Exp:8 Alpha-Beta Pruning Algorithm

Program:

```
def alpha_beta(depth, node_index, is_max, values, alpha, beta, max_depth):  
    if depth == max_depth:  
        return values[node_index]  
    if is_max:  
        best = float('-inf')  
        for i in range(2):  
            value = alpha_beta(depth + 1, node_index * 2 + i,  
                                False, values, alpha, beta, max_depth)  
            best = max(best, value)  
            alpha = max(alpha, best)  
            if beta <= alpha:  
                break  
        return best  
    else:  
        best = float('inf')  
        for i in range(2):  
            value = alpha_beta(depth + 1, node_index * 2 + i,  
                                True, values, alpha, beta, max_depth)  
            best = min(best, value)  
            beta = min(beta, best)  
            if beta <= alpha:  
                break  
        return best  
values = [3, 5, 6, 9, 1, 2, 0, -1]  
max_depth = 3  
result = alpha_beta(0, 0, True, values,  
                    float('-inf'), float('inf'), max_depth)
```

```
print("Optimal Value:", result)
```

Output:



```
Exp-8 Alpha-Beta Pruning Algorithm program.py - C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp...
File Edit Format Run Options Window Help
# Alpha-Beta Pruning Algorithm

def alpha_beta(depth, node_index, is_max, values, alpha, beta, max_depth):
    if depth == max_depth:
        return values[node_index]

    if is_max:
        best = float('-inf')

        for i in range(2):
            value = alpha_beta(depth + 1, node_index * 2 + i,
                                False, values, alpha, beta, max_depth)
            best = max(best, value)
            alpha = max(alpha, best)

            if beta <= alpha:
                break

        return best

    else:
        best = float('inf')

        for i in range(2):
            value = alpha_beta(depth + 1, node_index * 2 + i,
                                True, values, alpha, beta, max_depth)
            best = min(best, value)
            beta = min(beta, best)

            if beta <= alpha:
                break

        return best

# Leaf node values
values = [3, 5, 6, 9, 1, 2, 0, -1]
max_depth = 3

result = alpha_beta(0, 0, True, values,
                    float('-inf'), float('inf'), max_depth)

print("Optimal Value:", result)
```

```
IDLE Shell 3.13.5
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-8 Alpha-Beta Pruning A
lgorithm/Exp-8 Alpha-Beta Pruning Algorithm program.py
Optimal Value: 5
>>>
```

Ln: 45 Col: 0

Ln: 6 Col: 0